A First Course in Causal Inference Ch 7

Kodai Minesaki

Faculty of Engineering, The University of Tokyo

October 12, 2026

Ch. 7 Matched-Pairs Experiment

Ch. 7 Matched-Pairs Experiment

- The matched-pairs experiment (MPE) is the most extreme version of the SRE with only one treated unit and one control unit within each stratum.
- It has its own estimation and inference strategy

7.1 Design of the experiment and potential outcomes

- Consider an experiment with $2n$ units.
- If we have predictive covariates to the outcomes, we can pair units based on the similarity of covariates.
- With many covariates, we can define pairwise distances between units and the form pairs based on these distances.
- We assume that the pairs are formed based on the covariates and the discuss the subsequent design and analysis issues.

7.1 Design of the experiment and potential outcomes

Setup

- Let (i, j) index the unit j in pair i, where $i = 1, \ldots, n$ and $j = 1, 2$.
- Unit (i, j) has potential outcomes $Y_{ij}(1)$ and $Y_{i,j}(0)$ under the treatment and control, respectively.
- Within each pair, we randomly assign one into the treatment group $(Z_i = 1)$ and the control group $(Z_i = 0)$.

7.1 Design of the experiment and potential outcomes

Definition 7.1 (MPE)

The treatment assignments satisfy

$$(Z_i)_{i=1}^n \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}\left(\tfrac{1}{2}\right).$$

The observed outcomes within pair i are

$$Y_{i1} = Z_i Y_{i1}(1) + (1 - Z_i) Y_{i1}(0) = \begin{cases} Y_{i1}(1), & \text{if } Z_i = 1, \\ Y_{i1}(0), & \text{if } Z_i = 0, \end{cases}$$

and

$$Y_{i2} = Z_i Y_{i2}(0) + (1 - Z_i) Y_{i2}(1) = \begin{cases} Y_{i2}(0), & \text{if } Z_i = 1, \\ Y_{i2}(1), & \text{if } Z_i = 0. \end{cases}$$

So the observed data are $(Z_i, Y_{i1}, Y_{i2})_{i=1}^n$

7.3 Neymanian inference

- The average causal effect within pair i is

$$\tau_i = \frac{1}{2}\left[\{Y_{i1}(1) - Y_{i2}(0)\} + \{Y_{i2}(1) - Y_{i1}(0)\}\right].$$

- and the average causal effect for all units is

$$\tau = \frac{1}{n}\sum_{i=1}^{n}\tau_i = \frac{1}{2n}\sum_{i=1}^{n}\sum_{j=1}^{2}\{Y_{ij}(1) - Y_{ij}(0)\}.$$

- Both $\hat{\tau}_i$ and $\hat{\tau}$ are unbiased for τ_i and τ, respectively.

7.3 Neymanian inference

We can calculate the variance of $\hat{\tau}$.
For simplicity, let $A_i = Y_{i1}(1) - Y_{i2}(0)$ and $B_i = Y_{i2}(1) - Y_{i1}(0)$. Then

$$\hat{\tau}_i = Z_i A_i + (1 - Z_i) B_i = Z_i(A_i - B_i) + B_i.$$

Check the unbiasedness of $\hat{\tau}_i$:

$$\mathbb{E}[\hat{\tau}_i] = \mathbb{E}[Z_i](A_i - B_i) + B_i = \frac{1}{2}(A_i - B_i) + B_i = \frac{A_i + B_i}{2}. \quad (\because Z_i \sim \text{Bernoulli}(1/2).)$$

Here, $\mathrm{var}(\hat{\tau}_i) = \mathrm{var}(Z_i(A_i - B_i) + B_i) = \mathrm{var}(Z_i)(A_i - B_i)^2 = \frac{1}{4}(A_i - B_i)^2$.
Therefore,

$$\mathrm{var}(\hat{\tau}) = \mathrm{var}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{var}(\hat{\tau}_i)\right) = \frac{1}{4n^2}\sum_{i=1}^{n}\{Y_{i1}(1) - Y_{i2}(0) + Y_{i1}(0) - Y_{i2}(1)\}. \quad \square$$

7.3 Neymanian inference

- We cannot follow the same strategy under the SRE to estimate the variance of $\hat{\tau}$ under the MPE.
- In MPE, the within-pair sample variances of the outcomes are not well defined.
  - In SRE, we calculate variances for a treatment group and a control group in each stratum but in MPE, there is only one in both groups in each stratum.
- Is it possible to estimate the variance of $\hat{\tau}$ in the MPE?

7.3 Neymanian inference

- Think about the classic i.i.d. sampling:
- Suppose the $\hat{\tau}_i$'s are i.i.d. with mean μ and σ^2.
- Then the variance of $\hat{\tau} = n^{-1}\sum_{i=1}^{n}\hat{\tau}_i$ is σ^2/n.
- An unbiased estimator for σ^2 is $(n-1)^{-1}\sum_{i=1}^{n}(\hat{\tau}_i - \hat{\tau})^2$.
- So the unbiased estimator for $\mathrm{var}(\hat{\tau})$ is

$$\widehat{V} = \frac{1}{n(n-1)}\sum_{i=1}^{n}(\hat{\tau}_i - \hat{\tau})^2. \tag{1}$$

- We derive $\widehat{V}$ from a completely different setting from the MPE and a different motivation, so does it work for the MPE?

7.3 Neymanian inference

Theorem 7.1

Under the MPE, $\widehat{V}$ defined in (1) is a conservative estimator for the true variance of $\hat{\tau}$:

$$\mathbb{E}[\widehat{V}] - \mathrm{var}(\hat{\tau}) = \frac{1}{n(n-1)} \sum_{i=1}^{n} (\tau_i - \tau)^2 \geq 0.$$

If the τ_i's are constant across pairs, then $\mathbb{E}[\widehat{V}] = \mathrm{var}(\hat{\tau})$.

- Thm. 7.1 states that under the MPE,
  1. $\widehat{V}$ is a conservative variance estimator in general;
  2. $\widehat{V}$ is unbiased if the average causal effecsts are constant across pairs.

7.3 Neymanian inference

Proof of Thm. 7.1:
Recall $\sum_{i=1}^n (a_i - \bar{a})^2 = \sum_{i=1}^n a_i^2 - n\bar{a}^2$ and $\mathrm{var}[W] = \mathbb{E}[W^2] - (\mathbb{E}[W])^2$. Use them in the following manipulation:

$$
\begin{aligned}
n(n-1)\mathbb{E}[\widehat{V}] &= \mathbb{E}\left[\sum_{i=1}^n (\hat{\tau}_i - \hat{\tau})^2\right] = \mathbb{E}\left[\sum_{i=1}^n \hat{\tau}_i^2 - n\hat{\tau}^2\right] \\
&= \sum_{i=1}^n \{\mathrm{var}(\hat{\tau}) + \tau_i^2\} - n\{\mathrm{var}(\hat{\tau}) + \tau^2\} \\
&= n^2 \,\mathrm{var}(\hat{\tau}) - n\,\mathrm{var}(\hat{\tau}) + \sum_{i=1}^n (\tau_i - \tau)^2.
\end{aligned}
$$

Therefore,

$$
\mathbb{E}[\widehat{V}] = \mathrm{var}(\hat{\tau}) + \{n(n-1)\}^{-1} \sum_{i=1}^n (\tau_i - \tau)^2 \geq 0. \quad \square
$$

7.3 Neymanian inference

Both the point estimator $\hat{\tau}$ and the variance estimator $\widehat{V}$ can be conveniently obtained by OLS:

Proposition 7.1

$\hat{\tau}$ and $\widehat{V}$ are identical to the coefficient and variance estimator of the intercept from the OLS fit of the vector $(\hat{\tau}_1, \hat{\tau}_2, \ldots, \hat{\tau}_n)^T$ on the intercept only.

7.3 Neymanian inference

Proof of Prop. 7.1.

7.4 Covariate adjustment

- Although we have matched on covariates in the design stage, it is possible that the matching is not perfect $(X_{i1} \neq X_{i2})$.
- In this case, we can adjust for the covariates to further improve estimation efficiency.
- Assume that each unit (i, j) has covariates X_{ij}.

7.4 Covariate adjustment

- Define the within-pair differences in covariates:

$$\hat{\tau}_{X,i} = Z_i(X_{i1} - X_{i2}) + (1 - Z_i)(X_{i2} - X_{i1}) = (2Z_i - 1)(X_{i1} - X_{i2}),$$

- and their average:

$$\hat{\tau}_X = \frac{1}{n}\sum_{i=1}^{n} \hat{\tau}_{X,i}.$$

- We can show that

$$\mathbb{E}[\hat{\tau}_{X,i}] = 0, \quad \mathbb{E}[\hat{\tau}_X] = 0,$$

- and

$$\mathrm{cov}(\hat{\tau}_X) = \frac{1}{n^2}\sum_{i=1}^{n} \hat{\tau}_{X,i}\hat{\tau}_{X,i}^T.$$

7.4 Covariate adjustment

- In a realized MPE, $\mathrm{cov}(\hat{\tau}_X)$ is not zero unless all the $\hat{\tau}_X$'s are zero.
  $\rightsquigarrow$ Some realized values of $\hat{\tau}_X \neq 0$.
- This is the reason why there exists imbalance and we would like to adjust the imbalance to improve estimation efficiency.
- Similar to the discusion in the last chapter, consider a class of estimators:

$$\hat{\tau}_\gamma = \hat{\tau} - \gamma^T \hat{\tau}_X,$$

  which $\hat{\tau}_\gamma - \hat{\tau}$ has mean zero for any fixed γ.
- We want to choose γ to minimize the variance of $\hat{\tau}(\gamma)$.

7.4 Covariate adjustment

- The variance of $\hat{\tau}(\gamma)$ is

$$\begin{aligned}\text{var} &= \text{var}(\hat{\tau} - \gamma^T \hat{\tau}_X) \\ &= \text{var}(\hat{\tau}) + \gamma^T \text{cov}(\hat{\tau}_X)\gamma - 2\gamma^T \text{cov}(\hat{\tau}_X, \hat{\tau}),\end{aligned}$$

  which is minimized at

$$\tilde{\gamma} = \text{cov}(\hat{\tau}_X)^{-1} \text{cov}(\hat{\tau}_X, \hat{\tau}).$$

- Rewrite the formula for $\text{cov}(\hat{\tau}_X)$ as

$$\text{cov}(\hat{\tau}_X) = \frac{1}{n^2} \sum_{i=1}^{n} |\hat{\tau}_{X,i}||\hat{\tau}_{X,i}|^T,$$

  where $|\cdot|$ denotes the component-wise absolute value of a vector.

7.4 Covariate adjustment

- We can calculate $\mathrm{cov}(\hat{\tau}_X)$ but we cannot calculate $\mathrm{cov}(\hat{\tau}_X, \hat{\tau})$
  - This is because we need unknown potential outcomes to compute this.
- Fortunately, we can obtain an unbiased estimator for it.

Theorem 7.2

An unbiased estimator for $\mathrm{cov}(\hat{\tau}_X, \hat{\tau})$ is

$$\hat{\theta} = \frac{1}{n(n-1)} \sum_{i=1}^{n} (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau}).$$

7.4 Covariate adjustment

Substiting $\hat{\theta}$ into $\tilde{\gamma}$, then we obtain $\hat{\gamma}$ as follows:

$$\begin{aligned}
\hat{\gamma} &= \left(\frac{1}{n^2} \sum_{i=1}^{n} \hat{\tau}_{X,i} \hat{\tau}_{X,i}^T \right)^{-1} \left\{ \frac{1}{n(n-1)} \sum_{i=1}^{n} (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau}) \right\} \\
&= \left(\frac{1}{n^2} \sum_{i=1}^{n} (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_{X,i} - \hat{\tau}_X)^T + \frac{1}{n^2} \cdot n \hat{\tau}_X \hat{\tau}_X^T \right)^{-1} \left\{ \frac{1}{n(n-1)} \sum_{i=1}^{n} (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau}) \right\} \\
&\approx \left(\sum_{i=1}^{n} (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_{X,i} - \hat{\tau}_X)^T \right)^{-1} \left\{ (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau}) \right\}.
\end{aligned}$$

The second equality follows from $\sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$. The approximation holds from $\frac{n^2}{n(n-1)} \xrightarrow{n \to \infty} 1$ and $\frac{n \hat{\tau}_X \hat{\tau}_X^T}{n^2} \xrightarrow{n \to \infty} 0$.

7.4 Covariate adjustment

- This is approximately the coeeficient of the $\hat{\tau}_{X,i}$ in the OLS fit of the $\hat{\tau}_i$'s on the $\hat{\tau}_{X,i}$'s with an intercept.
- The final estimator is

$$\hat{\tau}_{\text{adj}} = \hat{\tau} - \hat{\gamma}^T \hat{\tau}_X,$$

  which is approximately the intercept in the OLS fit of the $\hat{\tau}_i$'s on the $\hat{\tau}_{X,i}$'s with an intercept.
- The conservative variance estimator for $\hat{\tau}_{\text{adj}}$ is then

$$\begin{aligned}
\widehat{V}_{\text{adj}} &= \text{var}(\hat{\tau} - \hat{\gamma}^T \hat{\tau}_X) \\
&= \widehat{V} + \hat{\gamma}^T \text{cov}(\hat{\tau}_X)\hat{\gamma} - 2\hat{\gamma}^T \hat{\theta} \\
&= \widehat{V} - \hat{\theta}\, \text{cov}(\hat{\tau}_X)^{-1} \hat{\theta}.
\end{aligned}$$

7.4 Covariate adjustment

Proposition 7.2

Under the MPE, the covariance-adjusted estimator $\hat{\tau}_{\text{adj}}$ and the associated variance estimator $\widehat{V}_{\text{adj}}$ can be conviniently approximated by the intercept and the associated variance estimator from the OLS fit of the vector of the $\hat{\tau}_i$'s on the 1's and the matrix of the $\hat{\tau}_{X,i}$'s.

- Neither Prop. 7.1 nor 7.2 requires the EHW correction of the variance estimator.
- In the MPE, we only use a pair so we do not need to center the covariate.
- Explicitly, let $\overline{X}$ be the mean covariate, then $X_{i1} - X_{i2} = (X_{i1} - \overline{X}) - (X_{i2} - \overline{X})$.

7.6 Comparing the MPE and CRE.

- Imai (2008) compared the MPE and CRE: the MPE gives more precise estimators if the matching is well done and the covariates are predictive of the outcomes.
- Without the outcome data in the design stafe, it is hard to decide whether this is true or not.
- Usually the MPE improves efficiency, it can be not true with small sample.
- $\rightsquigarrow$ The MPE is not superior to the CRE based on the FRT.

7.7 Extension to the general matched experiment

- Extend the MPE to the general matched experiment with varying numbers of control units.
  - If we consider a case with varying numbers of treatment units, it is essentially same to the case with varying numbers of control units.
  - If we consider a case with varying numbers of both treatment and control units, it is equivalent to the SRE.
- Namely, MPE $\subset$ Extended MPE $\subset$ SRE.

7.7 Extension to the general matched experiment

Setup

- n matched sets indexed by $i = 1, \ldots, n$.
- For the matched set i, $M_i + 1$ units; M_i's can vary.
- Total number of experimental units is $N = n + \sum_{i=1}^{n} M_i$.
- Let ij index the unit j within matched set i ($i = 1, \ldots, n$ and $j = 1, \ldots, M_i + 1$).
- Unit ij has potential outcomes $Y_{ij}(1)$ and $Y_{ij}(0)$.
- Within set i, 1 unit receives the treatment; the rest M_i recerives the control.
- Z_{ij}: the treatment indicator for unit ij.

7.7 Extension to the general matched experiment

- One of the potential outcomes:

$$Y_{ij} = Z_{ij} Y_{ij}(1) + (1 - Z_{ij}) Y_{ij}(0).$$

- The average causal effect within the matched set i equals:

$$\tau_i = \frac{1}{M_i + 1} \sum_{j=1}^{M_i+1} \{ Y_{ij}(1) - Y_{ij}(0) \}.$$

- An unbised estimator of τ_i is

$$\hat{\tau}_i = \sum_{j=1}^{M_i+1} Z_{ij} Y_{ij} - \frac{1}{M_i} \sum_{j=1}^{M_i+1} (1 - Z_{ij}) Y_{ij}.$$

7.7 Extension to the general matched experiment

- We first focus on estimating the average of the within-strata effects: $\tau = n^{-1} \sum_{i=1}^{n} \tau_i$.
- It has an unbiased estimator: $\hat{\tau} = n^{-1} \sum_{i=1}^{n} \hat{\tau}_i$.
- Importtantly, the τ above is the average of the τ_i's and it does not equal the average causal effect for the N units in the experiment when the M_i's vary.
- The average causal effect equals

$$\tau' = \frac{1}{N} \sum_{i=1}^{n} \sum_{j=1}^{1+M_i} \{Y_{ij}(1) - Y_{ij}(0)\} = \sum_{i=1}^{n} \frac{1+M_i}{N} \tau_i.$$

7.7 Extension to the general matched experiment

- To unify the discussion, we consider the weighted causal effect

$$\tau_w = \sum_{i=1}^{n} w_i \tau_i,$$

  with $\sum_{i=1}^{n} w_i = 1$.
- It is straightforward to obtain an unbiased estimator $\hat{\tau}_w = \sum_{i=1}^{n} w_i \hat{\tau}_i$, and calculate its variance $\mathrm{var}(\hat{\tau}_w) = \sum_{i=1}^{n} w_i^2 \, \mathrm{var}(\hat{\tau}_i)$.
- However, we have only one realized value of $\hat{\tau}_i$ so we cannot calculate $\mathrm{var}(\hat{\tau}_i)$.

7.7 Extension to the general matched experiment

- We give a final form of the variance estimator:

$$\hat{V}_w = \sum_{i=1}^{n} c_i(\hat{\tau}_i - \hat{\tau}_w)^2$$

where

$$c_i = \frac{\frac{w_i^2}{1-2w_i}}{1 + \sum_{i=1}^{n} \frac{w_i^2}{1-2w_i}}.$$

Theorem

Under the general matched experiment with varying M_i's, if $w_i < 1/2$ for all i's, then

$$\mathbb{E}[\hat{V}_w] - \mathrm{var}(\hat{\tau}/w) = \sum_{i=1}^{n} c_i(\tau_i - \tau_w)^2 \geq \mathrm{var}(\hat{\tau}_w) \geq 0,$$

with equality holding if the τ_i's are constant.

References I

Imai, K. (2008). Variance identification and efficiency analysis in randomized experiments under the matched-pair design. *Statistics in medicine 27*(24), 4857–4873.